22GE004-BASICS OF ELECTRONICS ENGINEERING

UNIT II & LP7-SIGNAL CONDITIONING USING DIODE

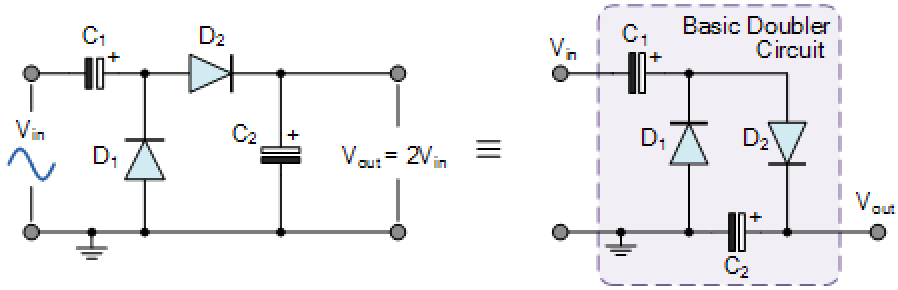
1. SIGNAL MULTIPLICATION USING PN JUNCTION DIODE

Voltage multipliers are similar in many ways to rectifiers in that they convert AC-to-DC voltages for use in many electrical and electronic circuit applications such as in microwave ovens, strong electric field coils for cathode-ray tubes, electrostatic and high voltage test equipment, etc, where it is necessary to have a very high DC voltage generated from a relatively low AC supply.

Generally, the DC output voltage (Vdc) of a rectifier circuit is limited by the peak value of its sinusoidal input voltage. But by using combinations of rectifier diodes and capacitors together we can effectively multiply this input peak voltage to give a DC output equal to some odd or even multiple of the peak voltage value of the AC input voltage.

1.1. The Voltage Doubler

As its name suggests, a Voltage Doubler is a voltage multiplier circuit which has a voltage multiplication factor of two. The circuit consists of only two diodes, two capacitors and an oscillating AC input voltage (a PWM waveform could also be used). This simple diode-capacitor pump circuit gives a DC output voltage equal to the peak-to-peak value of the sinusoidal input. In other words, double the peak voltage value because the diodes and the capacitors work together to effectively double the voltage.



DC VOLTAGE DOUBLER CIRCUIT

The circuit shows a half wave voltage doubler. During the negative half cycle of the sinusoidal input waveform, diode D1 is forward biased and conducts charging up the pump capacitor, C1 to the peak value of the input voltage, (Vp). Because there is no return path for capacitor C1 to discharge into, it remains fully charged acting as a storage device in series with the voltage supply. At the same time, diode D2 conducts via D1 charging up capacitor, C2.

During the positive half cycle, diode D1 is reverse biased blocking the discharging of C1 while diode D2 is forward biased charging up capacitor C2. But because there is a voltage across capacitor C1 already equal to the peak input voltage, capacitor C2 charges to twice the peak voltage value of the input signal.

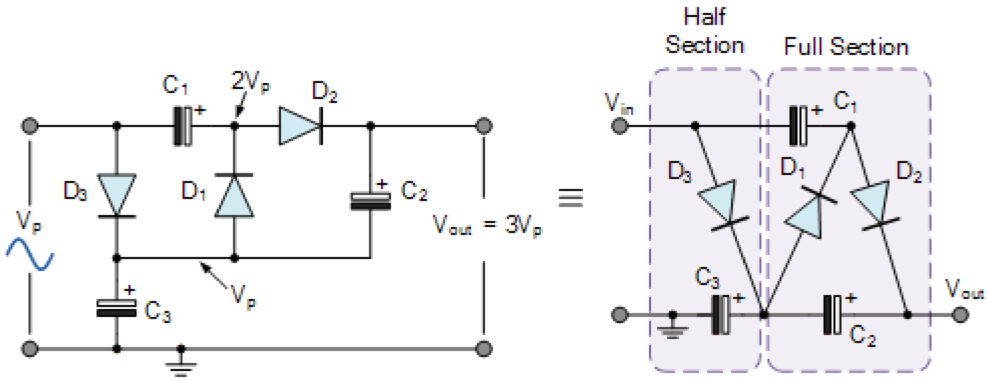
In other words, V(positive peak) + V(negative peak), so on the negative half-cycle, D1 charges C1 to Vp and on the positive half-cycle D2 adds the AC peak voltage to Vp onC1 and transfers it all to C2. The voltage across capacitor, C2 discharges through the load ready for the next half cycle.

Then the voltage across capacitor, C2 can be calculated as: Vout = 2Vp, (minus of course the voltage drops across the diodes used) where Vp is the peak value of the input voltage. Note that this double output voltage is not instantaneous but increases slowly on each input cycle, eventually settling to 2Vp.

As capacitor C2 only charges up during one half cycle of the input waveform, the resulting output voltage discharged into the load has a ripple frequency equal to the supply frequency, hence the name half wave voltage doubler. The disadvantage of this is that it can be difficult to smooth out this large ripple frequency in much the same way as for a half wave rectifier circuit. Also, capacitor C2 must have a DC voltage rating at least twice the value of the peak input voltage.

The advantage of Voltage Multiplier Circuits is that it allows higher voltages to be created from a low voltage power source without a need for an expensive high voltage transformer as the voltage doubler circuit makes it possible to use a transformer with a lower step up ratio than would be need if an ordinary full wave supply were used. However, while voltage multipliers can boost the voltage, they can only supply low currents to a high-resistance (+100kΩ) load because the generated output voltage quickly drops-off as load current increases.

1.2. DC Voltage Tripler Circuit



A VOLTAGE TRIPLER CIRCUIT

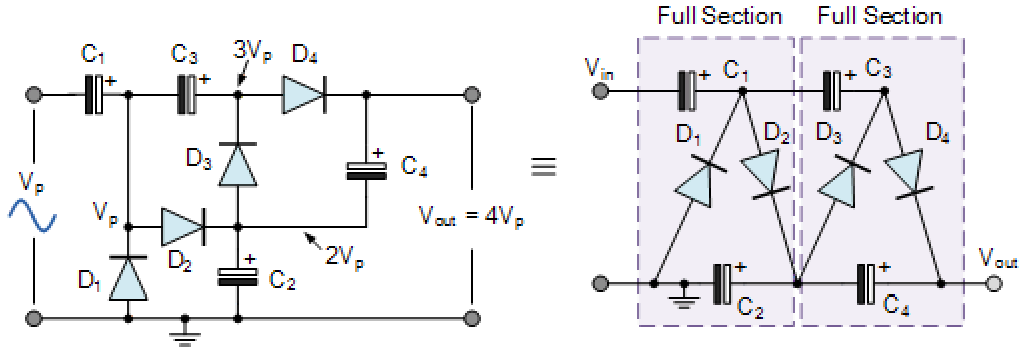
By adding an additional single diode-capacitor stage to the half-wave voltage doubler circuit above, we can create another voltage multiplier circuit that increases its input voltage by a factor of three and producing what is called a Voltage Tripler Circuit.

A “voltage tripler circuit” consists of one and a half voltage doubler stages. This voltage multiplier circuit gives a DC output equal to three times the peak voltage value (3Vp) of the sinusoidal input signal. As with the previous voltage doubler, the diodes within the voltage tripler circuit charge and block the discharge of the capacitors depending upon the direction of the input half-cycle. Then 1Vp is dropped across C3 and 2Vp across C2 and as the two capacitors are in series, this results in the load seeing a voltage equivalent to 3Vp.

Note that the real output voltage will be three times the peak input voltage minus the voltage drops across the diodes used, 3Vp – V(diode).

If a voltage tripler circuit can be made by cascading together one and a half voltage multipliers, then a Voltage Quadrupler Circuit can be constructed by cascading together two full voltage doubler circuits.

1.3. DC Voltage Quadrupler Circuit



The first voltage multiplier stage doubles the peak input voltage and the second stage doubles it again, giving a DC output equal to four times the peak voltage value (4Vp) of the sinusoidal input signal. Also, using large value capacitors will help to reduce the ripple voltage.